EXCELLENCE

CLASS 11 · 2026–27

Physics | Lecture - 01

Basic Maths and Calculus

Trigonometry

By — Excellence Programme

1 September 2026

Topics to be covered

01 Trigonometric ratios — sin, cos, tan, cosec, sec, cot

02 Trigonometric identities from $P^2 + B^2 = H^2$

03 Finding unknown sides from a given ratio

04 Range of sine and cosine

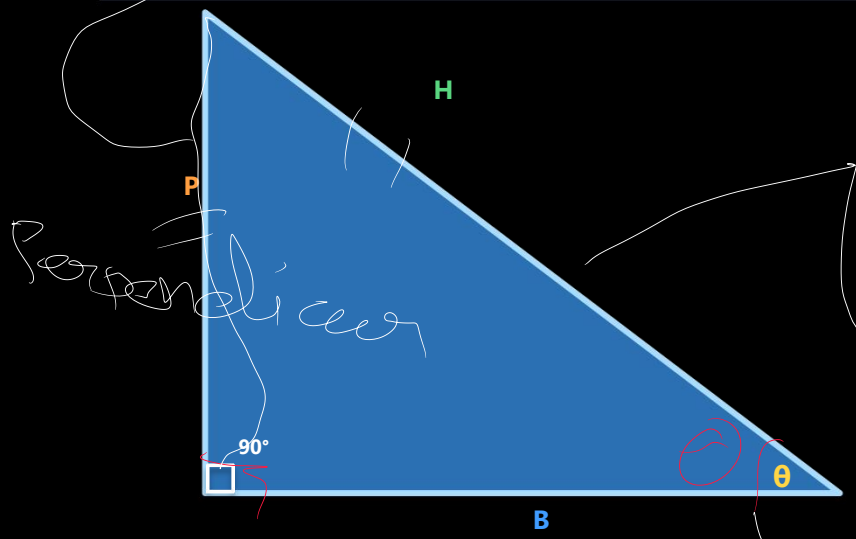
05 Algebra with ratios, and similar triangles

06 Height and distance

07 Standard angles 0° to 180°, and 37° / 53°

08 Radian measure, arc = Rθ, degree ↔ radian

Name the sides from θ



P — perpendicular — opposite θ
B — base — adjacent to θ
H — hypotenuse — longest side

Not from how the figure sits.

$$\sin\theta = \frac{P}{H}$$

$$\cos\theta = \frac{B}{H}$$

$$p^2 + b^2 = h^2$$

sin θ and cosec θ

sin θ = P / H

Opposite over hypotenuse.

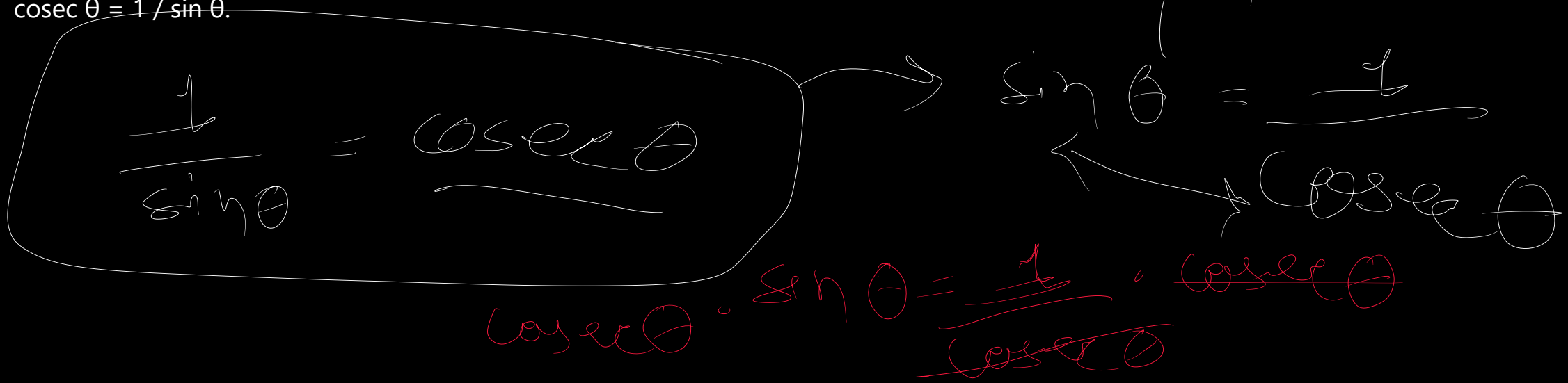
cosec θ = H / P

Reciprocal of sin θ. If sin θ = 3/5, then cosec θ = 5/3.

sin, cos, tan are dimensionless. Never write sin θ = 0.6 m.

cosec θ = 1 / sin θ.

sin θ · cosec θ = 1



cos θ and sec θ

cos θ = B / H

Adjacent over hypotenuse.

sec θ = H / B

Reciprocal of cos θ. If cos θ = 4/5, then sec θ = 5/4.

Never write cos θ = 0.8 m. A ratio has no unit.

sec θ = 1 / cos θ.

cos θ · sec θ = 1

$\cos\theta = \frac{1}{\sec\theta}$

$\cos\theta = \frac{B}{H} \rightarrow \sec\theta = \frac{H}{B} = \frac{1}{\frac{B}{H}}$

cos θ and sec θ

cos θ = B / H

Adjacent over hypotenuse.

sec θ = H / B

Reciprocal of cos θ. If cos θ = 4/5, then sec θ = 5/4.

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$\cos\theta = \frac{1}{\sec\theta}$

$\cos\theta = \frac{B}{H} \rightarrow \sec\theta = \frac{H}{B} = \frac{1}{\frac{B}{H}}$

tan θ and cot θ

tan θ = P / B

Opposite over adjacent.

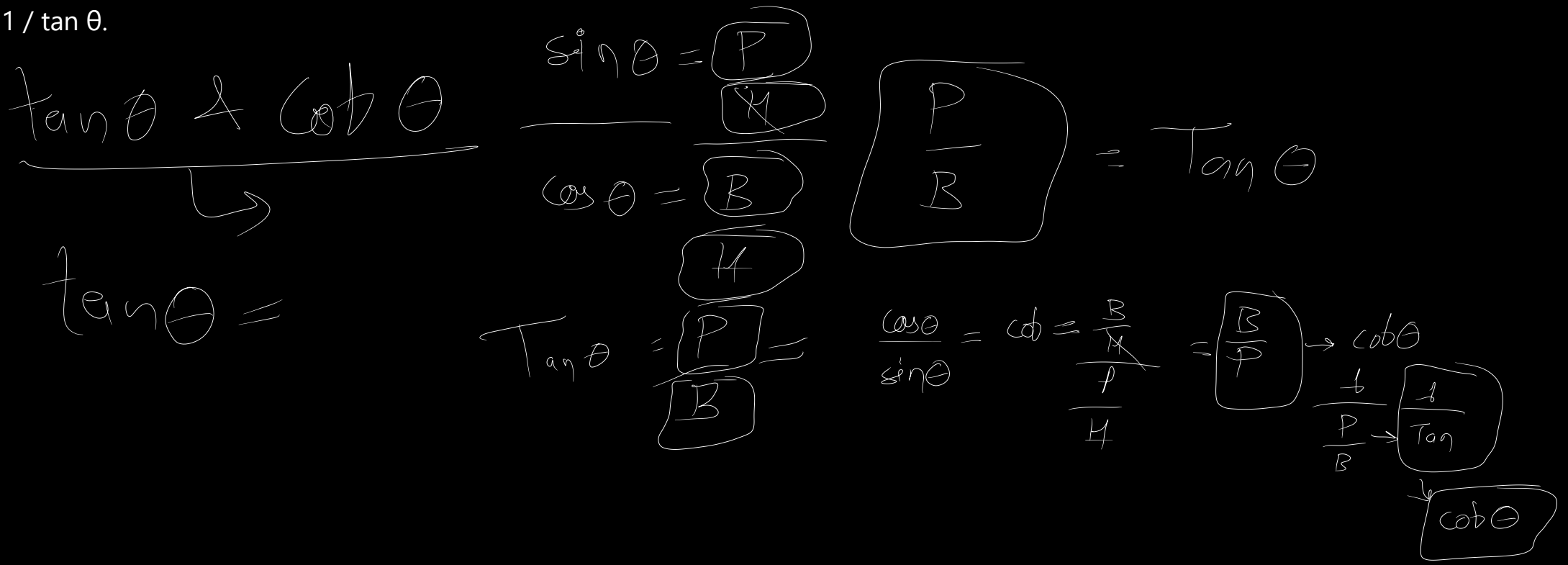
cot θ = B / P

Reciprocal of tan θ. Also: tan θ = sin θ / cos θ.

sin · cosec = 1 cos · sec = 1 tan · cot = 1

cot θ = 1 / tan θ.

tan θ · cot θ = 1



Divide Pythagoras by H^2

Start here. Divide both sides by H^2.

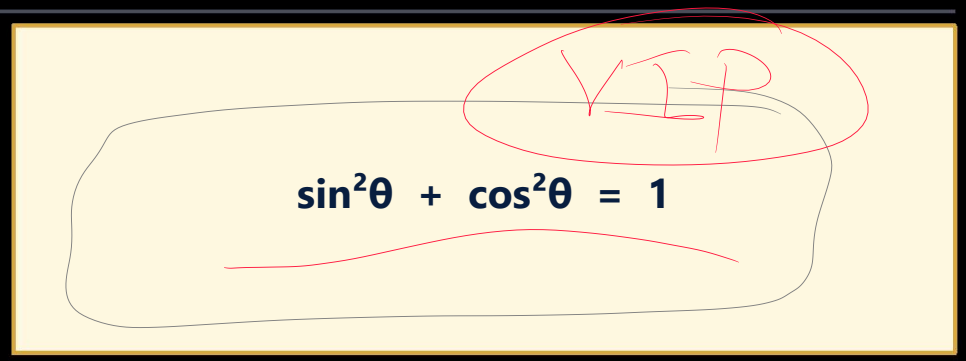
$$P^2 + B^2 = H^2$$

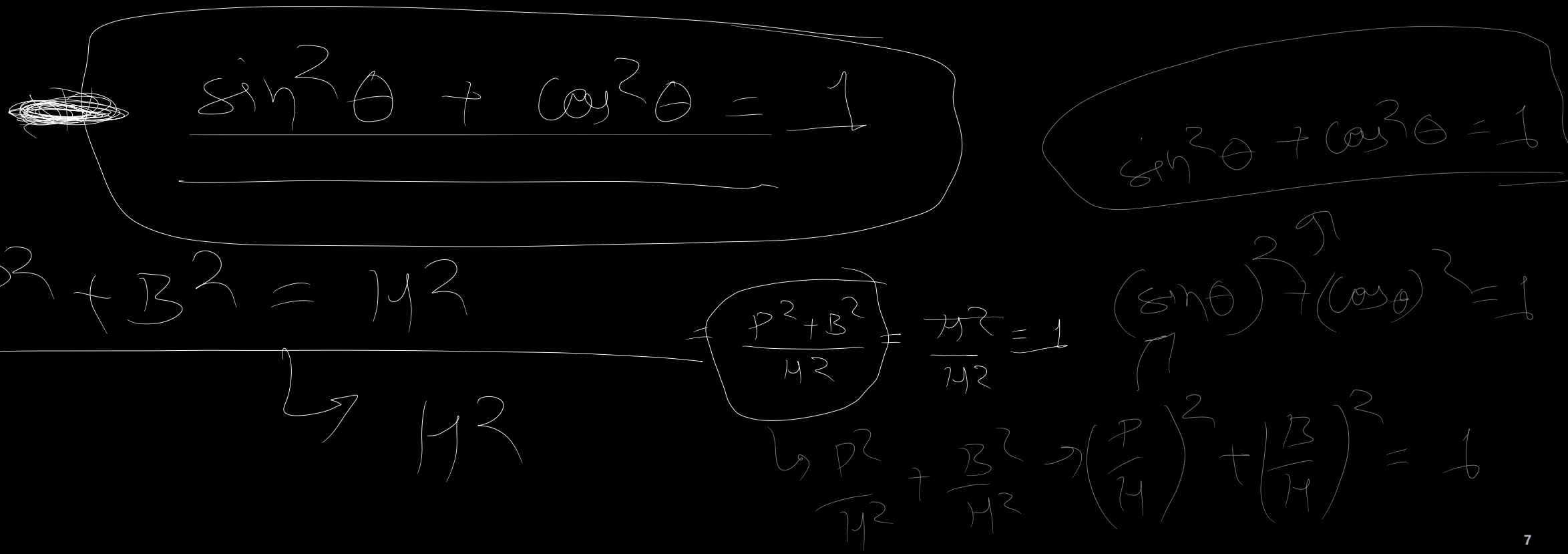
$$P^2 / H^2 + B^2 / H^2 = H^2 / H^2$$

$$(P / H)^2 + (B / H)^2 = 1$$

Also: $\cos\theta = \sqrt{(1 - \sin^2\theta)}$ and $\sin\theta = \sqrt{(1 - \cos^2\theta)}$

$$\sin^2\theta + \cos^2\theta = 1$$





Divide Pythagoras by H^2

Start here. Divide both sides by H^2.

$$P^2 + B^2 = H^2$$

$$P^2 / H^2 + B^2 / H^2 = H^2 / H^2$$

$$(P / H)^2 + (B / H)^2 = 1$$

Also: $\cos\theta = \sqrt{(1 - \sin^2\theta)}$ and $\sin\theta = \sqrt{(1 - \cos^2\theta)}$

$$\sin^2\theta = 1 - \cos^2\theta$$

$$\sin\theta = \sqrt{1 - \cos^2\theta}$$

$$\sin^2\theta + \cos^2\theta = 1$$

Divide Pythagoras by B^2

Same start. Now divide both sides by B^2.

$$P^2 + B^2 = H^2$$

$$P^2 / B^2 + B^2 / B^2 = H^2 / B^2$$

$$(P / B)^2 + 1 = (H / B)^2$$

Because $P/B = \tan\theta$ and $H/B = \sec\theta$.

$$1 + \tan^2\theta = \sec^2\theta$$

Divide Pythagoras by P^2

Same start. Now divide both sides by P^2.

$$P^2 + B^2 = H^2$$

$$P^2 / P^2 + B^2 / P^2 = H^2 / P^2$$

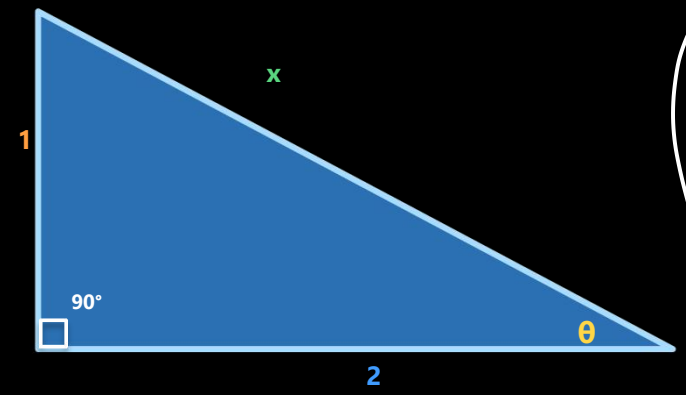
$$1 + (B / P)^2 = (H / P)^2$$

Because $B/P = \cot\theta$ and $H/P = \operatorname{cosec}\theta$. If you keep only one line: $\sin^2\theta + \cos^2\theta = 1$

$$1 + \cot^2\theta = \operatorname{cosec}^2\theta$$

Find x

In the right triangle shown, the perpendicular is 1 and the base is 2. The hypotenuse is x. Find the value of x, sin θ, cos θ and tan θ.



P = 1

B = 2

H = ? = x

$P^2 + B^2 = H^2$

$1^2 + 2^2 = H^2$

$1 + 4 = H^2 \rightarrow$

$H^2 = 5$

$H = \sqrt{5}$

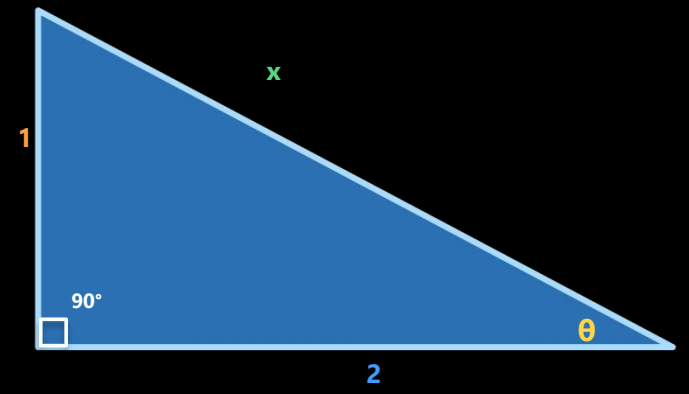
$\sin\theta = \frac{1}{\sqrt{5}}$

$\cos\theta = ?$

$= \frac{2}{\sqrt{5}}$

First find H

Right triangle: P = 1, B = 2, H = x.



$P^2 + B^2 = H^2$

$1^2 + 2^2 = x^2$

$x = \sqrt{5}$

Now sin θ, cos θ, tan θ

P = 1, B = 2, H = √5

sin θ = P / H = 1 / √5
cos θ = B / H = 2 / √5
tan θ = P / B = 1 / 2

Check: (1/√5) / (2/√5) = 1/2. Same.

tan θ = sin θ / cos θ = 1/2

$$\tan = \frac{P}{B} = \frac{1}{2}$$

If tan θ = 2

$P = x$

If tan θ = 2, then find the perpendicular, the base and the hypotenuse. Also find sin θ and cos θ. Take θ acute.

$\tan = 2 = \frac{P}{B} \rightarrow \frac{x}{B} = 2$

$x = 2B$

$x = 1 \quad B = 2$

$x = 2 \rightarrow B = 4$

$P = 4 \rightarrow B = 8$

$H^2 = P^2 + B^2 = 4^2 + 8^2 = 16 + 64 \Rightarrow 80$

$H^2 = 80 \rightarrow$

$H = \sqrt{80}$

$H = \sqrt{2 \times 2 \times 2 \times 2 \times 5}$

$2 \times 2 \times \sqrt{5}$

$= 4\sqrt{5}$

$\sin\theta = ? = \frac{P}{H} \rightarrow$

$= \frac{4}{\sqrt{\quad}}$

$\sin\theta = \frac{x}{4\sqrt{5}} = \frac{1}{\sqrt{5}}$

$P = 1 \rightarrow \frac{1}{\sqrt{3}}$

$P = 1 \rightarrow B = 2 \rightarrow H^2 = 1 + 4 = 5$

$\hookrightarrow \sqrt{5} = H$

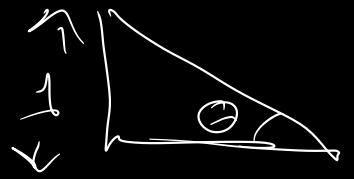
$H = \sqrt{5}$

$\sin\theta = \frac{1}{\sqrt{5}}$

If tan θ = 2

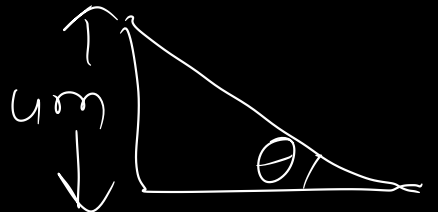
If tan θ = 2, then find the perpendicular, the base and the hypotenuse. Also find sin θ and cos θ. Take θ acute.

P = 1 →



$\sin\theta = \frac{1}{\sqrt{5}}$

P = 4



$\sin\theta = \frac{4}{\sqrt{5}}$

If θ is same ratio side of 2 Triangle is always Equal

SOLUTION

$\tan\theta = 2 \rightarrow$ find P, B, H

Given $\tan\theta = 2$ (θ acute).

$$\tan\theta = P/B = 2 = 2/1$$

Let B = k. Then P = 2k.

$$H^2 = (2k)^2 + k^2 = 5k^2$$

$H = \sqrt{5}\,k$ (take + because θ is acute)

P = 2k B = k H = √5 k

Now sin θ and cos θ

P = 2k, B = k, H = √5 k

sin θ = P / H = 2k / √5 k = 2 / √5

k cancels.

cos θ = B / H = k / √5 k = 1 / √5

k cancels.

A ratio only fixes the shape. The scale k always cancels in sin, cos, tan.

Range of sin θ

If sin θ = 2, then find the value of cos θ and tan θ.

sin θ → Range of sin θ is always in between

−1 & 1

$-1 \le \sin\theta, \cos\theta \le 1$

VIP

$\frac{\sqrt{-12}}{2} \rightarrow \frac{\sqrt{-3}}{\sqrt{}}$

Method 1 — geometry

If sin θ = 2, find cos θ and tan θ.

Wrong: Start calculating cos θ from the identity without checking the range.

Right: First ask: can sin θ equal 2?

→ first check Range

sin θ = P / H

H is the longest side, so H > P. Then P/H ≤ 1.

sin θ = 2 is impossible. There is no such triangle.



$\sin\theta = 2 \Rightarrow \frac{P}{H}$ → $P = 2H$

P can't be twice of H

→ H is longest side

Method 2 — identity



Try the identity anyway. It also fails.

$\sin^2\theta + \cos^2\theta = 1$

Put $\sin\theta = 2$.

$(2)^2 + \cos^2\theta = 1$

$\cos^2\theta = 1 - 4 = -3$. Not a real value.

$-1 \le \sin\theta \le 1$ and $-1 \le \cos\theta \le 1$ $\tan\theta$ has no finite range.

Find tan θ from a sin–cos fraction

M. W

If (sin θ + cos θ) / (sin θ − cos θ) = 11/5 , then find the value of tan θ.

$$\frac{\sin\theta + \cos\theta}{\sin\theta - \cos\theta} = \frac{11}{5}$$

$\Rightarrow 5(\sin\theta + \cos\theta) = 11(\sin\theta - \cos\theta)$

$= 5\sin\theta + 5\cos\theta = 11\sin\theta - 11\cos\theta$

$5\cos\theta + 11\cos\theta = 11\sin\theta - 5\sin\theta$

$16\cos\theta = 6\sin\theta$

$\frac{16}{6} = \frac{\sin\theta}{\cos\theta} =$

$\frac{8}{3} = \tan\theta$

$\tan\theta = ?$

$\tan\theta \Rightarrow \frac{8}{3}$

Cross multiply, then expand

$(\sin\theta + \cos\theta) / (\sin\theta - \cos\theta) = 11/5$

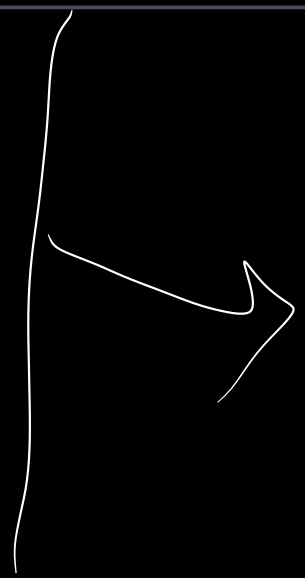
$5(\sin\theta + \cos\theta) = 11(\sin\theta - \cos\theta)$

Cross multiply.

$5\sin\theta + 5\cos\theta = 11\sin\theta - 11\cos\theta$

Open the brackets.

Next: collect sin on one side, cos on the other.



Collect, then divide

$5 \sin \theta + 5 \cos \theta = 11 \sin \theta - 11 \cos \theta$

$5 \cos \theta + 11 \cos \theta = 11 \sin \theta - 5 \sin \theta$

$16 \cos \theta = 6 \sin \theta$

$\sin \theta / \cos \theta = 16 / 6 = 8 / 3$

$\tan \theta = 8 / 3$

Nested right triangles — same θ, same tan θ

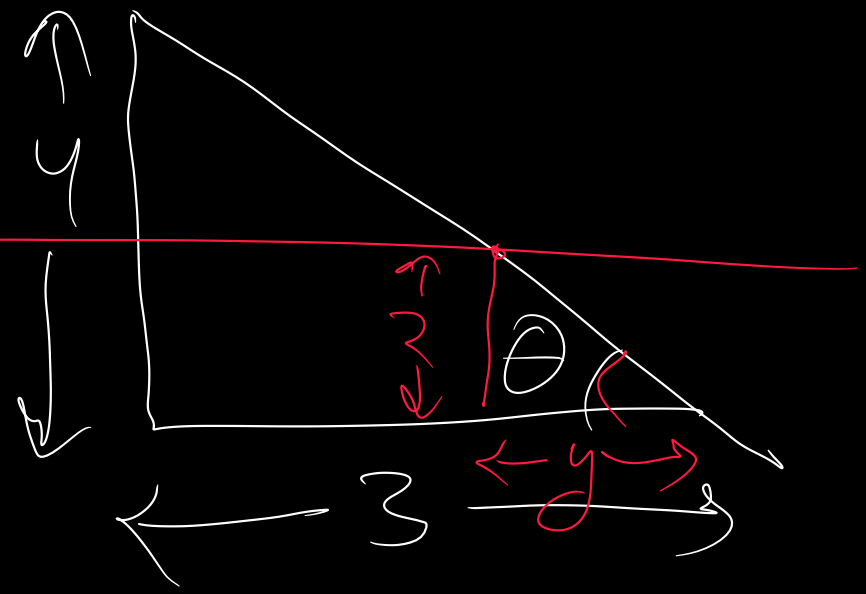
A large right triangle has opposite side 4 and adjacent side 3. Nested similar slices share the same angle θ. In the small slice the opposite is 2 and the adjacent is y. In the middle slice the opposite is x and the adjacent is 2. Find the values of x and y.

1. x = 4/3 , y = 3/2
2. x = 8/3 , y = 6/2
3. x = 8/3 , y = 3/2
4. x = 3/2 , y = 8/3

$\tan\theta = \frac{4}{3}$

$y = ?$

$\tan = \frac{P}{B} = \frac{4}{3}$



$\tan\theta = \frac{4}{3} = \frac{P}{B} = \frac{2}{y} \Rightarrow \frac{2}{y} = \frac{4}{3} \Rightarrow y = \frac{6}{4} = \frac{3}{2}$

$y = \frac{3}{2}$

Nested right triangles — same θ, same tan θ

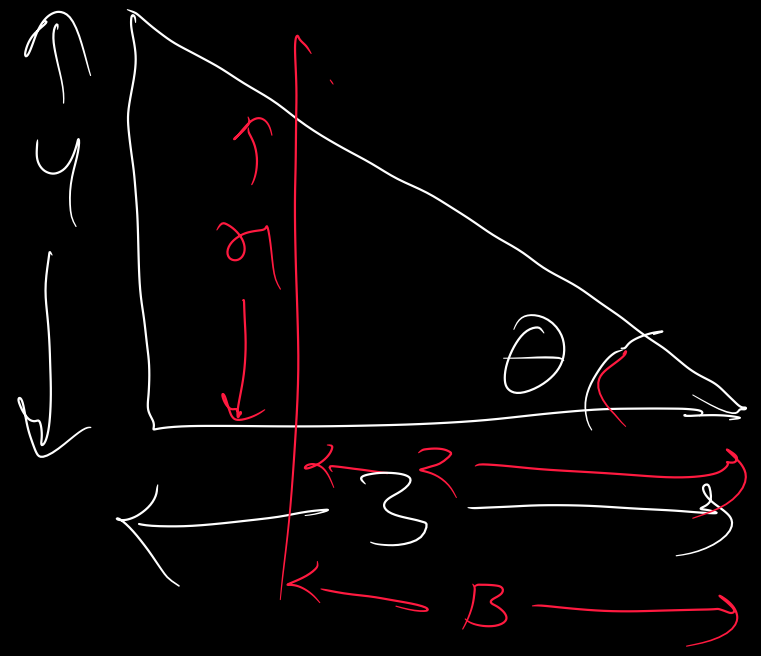
3

A large right triangle has opposite side 4 and adjacent side 3. Nested similar slices share the same angle θ. In the small slice the opposite is 2 and the adjacent is y. In the middle slice the opposite is x and the adjacent is 2. Find the values of x and y.

1. x = 4/3 , y = 3/2
2. x = 8/3 , y = 6/2
3. x = 8/3 , y = 3/2
4. x = 3/2 , y = 8/3

$x = ?$

$\tan\theta = \frac{4}{3} = \frac{x}{2}$



$x = \frac{8}{3}$ → $y = \frac{3}{2}$

for all nested △ $\tan\theta$, $\sin\theta$ & $\cos\theta$ is same

Find y from tan θ = 4/3

Large triangle: opposite 4, adjacent 3. Small slice: opposite 2, adjacent y.

tan θ = 4 / 3 = 2 / y

Same θ, so same tan.

4 y = 6

Cross multiply.

y = 3 / 2

Find x from tan θ = 4/3

Middle slice: opposite x, adjacent 2. Same θ as the large triangle.

tan θ = 4 / 3 = x / 2

Same θ, so same tan.

3 x = 8

Cross multiply.

x = 8 / 3

Do not mix a height from one triangle with a base from another. Each ratio must come from the SAME triangle.

BREAK

Next: height and distance → 0°–180° table → 37°/53° → radian

Observer between two towers — elevation

From a point on level ground between two vertical towers that are d metres apart, the angles of elevation to the tops of the nearer and farther towers are θ and φ respectively. The observer is x metres from the nearer tower. What is the difference in heights of the towers?

1. $d\,(\tan\theta - \tan\varphi)$
2. $x\tan\theta - (d - x)\tan\varphi$
3. $\tan\theta - \tan\varphi \cdot d$
4. $(\tan\theta - \tan\varphi) / d$

Nearer tower

Observer is x metres from the nearer tower. Elevation to that top is θ.

$$\tan\theta = h_1 / x$$

Opposite = height. Adjacent = x.

$$h_1 = x \tan\theta$$

Next: the farther tower.

Farther tower, then the difference

Observer is (d − x) metres from the farther tower. Elevation to that top is φ.

$$h_2 = (d - x) \tan \varphi$$

Same idea. Adjacent is now d − x.

$$h_1 - h_2 = x \tan \theta - (d - x) \tan \varphi$$

If the question asks for the farther minus the nearer, flip the sign. Always sketch before you write tan.

From the top of one tower — depression

Two towers are separated by a horizontal distance D. From the top of the first tower (height h_1), the angle of depression to the top of the second tower (height h_2) is θ. Which relation correctly gives D in terms of h_1, h_2 and θ? Take $h_1 > h_2$.

1. $D = (h_1 + h_2) / \tan\theta$
2. $D = (h_1 - h_2) / \tan\theta$
3. $D = (h_2 - h_1) / \tan\theta$
4. $D = \tan\theta \cdot (h_1 + h_2)$

Write tan θ first

From the top of the first tower, look down to the top of the second. $h_1 > h_2$.

$\tan\theta = (h_1 - h_2) / D$

Opposite = height drop. Adjacent = D.

Depression looks down, so the opposite side is the difference of heights, never the sum.

Next: solve for D.

Solve for D

$\tan\theta = (h_1 - h_2) / D$

$D = (h_1 - h_2) / \tan\theta$

Wrong options use $(h_1 + h_2)$ or multiply by $\tan\theta$. Depression looks down, so the opposite side is the difference of heights, never the sum.

sin, cos, tan from 0° to 90°

θ	0°	30°	45°	60°	90°
sin	0	1/2	1/√2	√3/2	1
cos	1	√3/2	1/√2	1/2	0
tan	0	1/√3	1	√3	undef.

tan 90° is not defined.

sin, cos, tan from 90° to 180°

θ	90°	120°	135°	150°	180°
sin	1	√3/2	1/√2	1/2	0
cos	0	−1/2	−1/√2	−√3/2	−1
tan	undef.	−√3	−1	−1/√3	0

Range: $-1 \le \sin\theta \le 1$ and $-1 \le \cos\theta \le 1$

Read these from the table

From the 0° to 180° table, find the value of each of the following.

(i) sin 120°
(ii) cos 150°
(iii) tan 135°
(iv) sin 150°
(v) cos 120°

From the 0°–180° table

$\sin 120° = \sqrt{3} / 2$

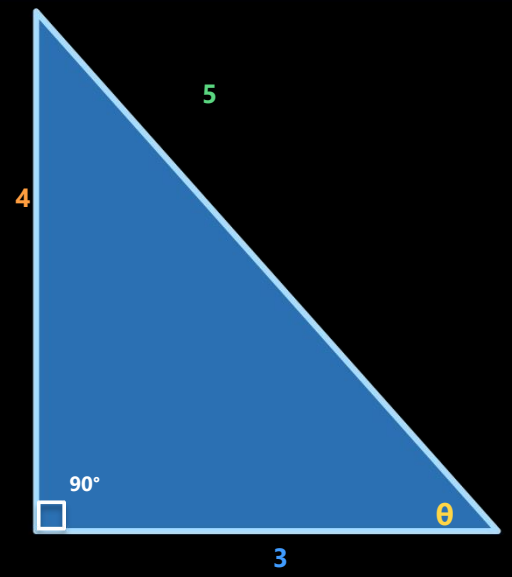
$\cos 150° = -\sqrt{3} / 2$

$\tan 135° = -1$

$\sin 150° = 1 / 2$

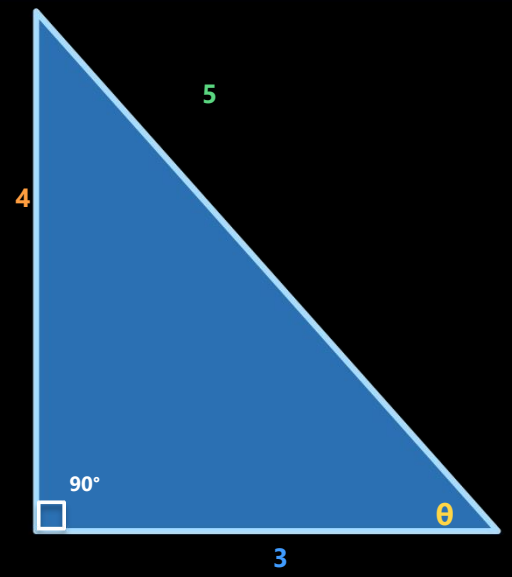
$\cos 120° = -1 / 2$

3–4–5 triangle — 37° and 53°



In a 3–4–5 right triangle, find sin, cos and tan of 37°, and sin, cos and tan of 53°.

3–4–5 triangle — 37° and 53°



sin 37° = 3/5 **cos 37° = 4/5** **tan 37° = 3/4**

sin 53° = 4/5 **cos 53° = 3/5** **tan 53° = 4/3**

tan 37° < tan 53° because 3/4 < 4/3.

Also: sin 53° × cosec 53° = 1.

From 0° to 90°

If the angle increases from 0° to 90°, then what happens to the value of sin θ and the value of cos θ?

1. sin θ increases, cos θ decreases
2. sin θ decreases, cos θ increases
3. both decrease
4. both increase

From 0° to 90°

sin θ increases

0 → 1/2 → 1/√2 → √3/2 → 1

cos θ decreases

1 → √3/2 → 1/√2 → 1/2 → 0

0° → 90°: sin ↑ cos ↓ tan ↑ (tan 90° not defined)

Identity

Find the value of $\sin^2 40° + \cos^2 40°$.

$\sin^2 40° + \cos^2 40°$

True for every θ. Here θ = 40°.

$$\sin^2\theta + \cos^2\theta = 1$$

Do not compute 40°. Same angle, so the identity is enough.

$$\sin^2 40° + \cos^2 40° = 1$$

Arc length

Arc length = R θ

θ must be in radian. Never put 90° into this formula.

Practical unit of angle: degree. SI unit of angle: radian.

+θ anti-clockwise −θ clockwise

Arc = R θ

The one line: 180° = π rad

Lock this. Every conversion starts here.

180° = π rad

Next: degree → radian, then radian → degree.

Wherever you see π rad, put 180°.

Degree → radian

1° = π / 180 rad

Divide the lock-line by 180.

So θ° in radian = θ × (π / 180).

Multiply the degree by π/180.

Radian → degree

---

1 rad = 180° / π

Divide the lock-line by π.

So θ rad in degree = θ × (180 / π).

Multiply the radian by 180/π.

Arc on a circle

An object moves on the circumference of a circle of radius R. If the object turns through 90°, find the total distance moved by the object.

Object turns 90° on a circle of radius R

An object moves on the circumference of a circle of radius R and turns through 90°. Find the distance.

Wrong: Distance = R × 90°

Right: Distance = R × (π/2 rad) = πR/2

distance = path = arc = Rθ

Convert 90° first: 90° = π/2 rad

πR / 2

First conversion

Convert π/3 radian into degree.

π / 3 rad

Wherever you see π rad, put 180°.

π / 3 rad = 180° / 3 = 60°

π / 3 rad = 60°

Convert these now

Convert the following into degree or radian, as required.

(i) π/2 rad

(ii) π/4 rad

(iii) π/6 rad

(iv) 45°

(v) 90°

(vi) 60°

The six conversions from the board

$\pi/2$ rad = 90°

$\pi/6$ rad = 30°

90° = $\pi/2$ rad

$\pi/4$ rad = 45°

45° = $\pi/4$ rad

60° = $\pi/3$ rad

Find β

A triangle has two interior angles 40° and 70°. The side next to the 70° angle is extended. If β is the exterior angle formed at that vertex, find the value of β.

Convert radian into degree

Convert the following radian measures into degree.

(i) $5\pi/6$

(ii) $2\pi/3$

(iii) $4\pi/3$

(iv) $3\pi/5$

(v) 5π

(vi) $2\pi/5$

(vii) $7\pi/12$

(viii) $5\pi/2$

(ix) 3π

Convert degree into radian

Convert the following into radian.

(i) 120°

(ii) 30°

(iii) 150°

(iv) 270°

(v) 135°

(vi) π

Product of a ratio and its reciprocal

Find the value of $\sin 53° \times \operatorname{cosec} 53°$.

Product of three values

Find the value of $\sin 180° \cdot \cos 135° \cdot \tan 120°$.

Shadow of a pole

A vertical pole of height h casts a shadow of length l when the Sun is at an angle θ above the horizon. Which of the following expresses the shadow length l in terms of h and θ?

1. $l = h \tan \theta$
2. $l = h / \tan \theta$
3. $l = h \sin \theta$
4. $l = h / \sin \theta$

Wire from the top of a pole

A pole of height h has a wire attached from the top to the ground at an angle θ. If the wire breaks and falls to the ground without changing length, find the horizontal distance x from the base of the pole where the wire touches the ground.

1. $x = h \cot \theta$
2. $x = h \tan \theta$
3. $x = h \sin \theta$
4. $x = h \sec \theta$

Shadow on sloping ground

A vertical pole of height $h = 10$ m stands on ground that slopes upwards at a constant angle $\alpha = 30°$ with the horizontal. If the Sun's angle of elevation above the horizontal is $\theta = 60°$, what is the length of the shadow cast by the pole on the sloping ground?

1. 5 m
2. 10 m
3. $10\sqrt{3}$ m
4. $10 / (\sqrt{3} - 1)$ m

After class

Daily test before the next lecture.

P, B, H

sin = P/H cos = B/H tan = P/B and the three reciprocals

Identities

$$\sin^2\theta + \cos^2\theta = 1 \qquad 1 + \tan^2\theta = \sec^2\theta \qquad 1 + \cot^2\theta = \operatorname{cosec}^2\theta$$

Range and table

$-1 \le \sin, \cos \le 1$ 0°–180° table tan 90° not defined

Radian

$\text{Arc} = R\theta$ with θ in radian π rad $\rightarrow 180°$

After class

Daily test tonight.

Remaining conversions and PYQs tonight.